

EGERTON



UNIVERSITY

UNIVERSITY EXAMINATIONS

REGULAR -NJORO CAMPUS

SECOND SEMESTER, 2021/2022 ACADEMIC YEAR

THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN

NATURAL RESOURCE MANAGEMENT

NARE 352: ECOLOGICAL SURVEYS AND TECHNIQUES

STREAM: BSc NARE, TOHM & ENSCI (Y3S2)

TIME: 2 HRS

EXAMINATION SESSION: JULY

YEAR: 2022

INSTRUCTIONS:

- (i) Answer ALL questions in Section A and any two in section B
- (ii) Some formulae are provided at the end of the question paper
- (iii) Do not write on the question paper

SECTION A (40 MARKS)

Question One

a) Define the following terms commonly used in ecological surveys and techniques:

- i). Sampling frame **(1 Mark)**
- ii). Basal cover **(1 Mark)**

b) Distinguish between the following:

- i). Habitat and habitat type **(2 Marks)**
- ii). Abstract and executive summary **(2 Marks)**

Question Two

Write short notes on the following:

- i). Remote sensing **(4 Marks)**
- ii). The "Discussion" section of an ecological monitoring report **(4 Marks)**
- iii). Systematic sampling **(4 Marks)**

Question Three

- a) Give three reasons why proper ecological database management is important (6 Marks)
- b) Outline six basic descriptors that should be contained ecological database. (6 Marks)

Question Four

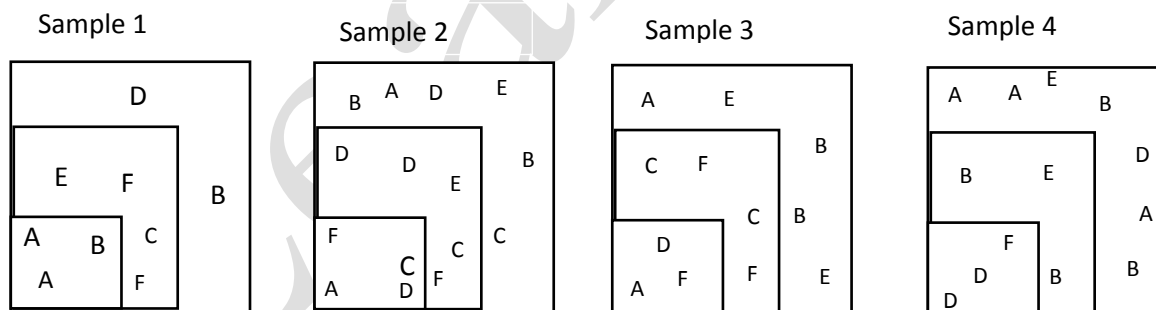
- a) Define reverse planning, and give four reasons for proper planning before embarking on an ecological survey. (6 Marks)
- b) Outline the steps for deciding whether data can be collected in the time. (4 Marks)

SECTION B (30 MARKS)**Question Five**

- a) Write short notes on plant density. (7 Marks)
- b) Describe how to estimate plant density using the PCQ method. (8 Marks)

Question Six

- a) The Figure below shows the location of individuals of various herbaceous plant species (A" F) within quadrats of different sizes (small, medium and large) placed in a nested design at four locations (Sample 1-4) across a study site. For each quadrat size, calculate the frequency of each species in the study site. (9 Marks)



- b) Outline four suitable situations for the method applied above (a). (6 Marks)

Question Seven

- a) Illustrate any four types of graphs commonly employed to present ecological data, and briefly explain the use each graph type. (8 Marks)
- b) You intend to estimate the population size of wild large herbivore species in a conservancy. Formulate a data collection protocol for this study. (7 Marks)
